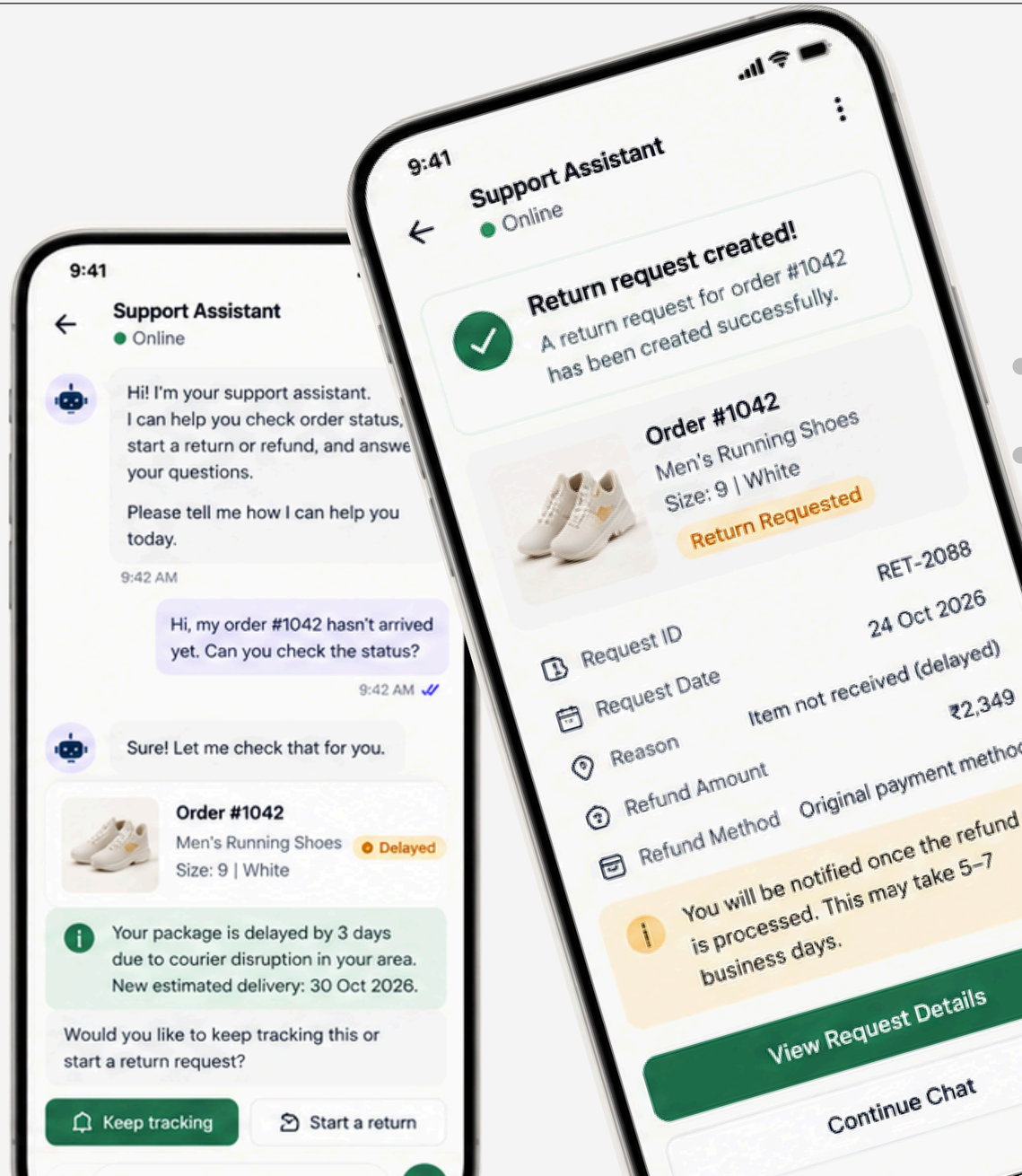
SellerResolve

PS04

Autonomous support agent for online sellers

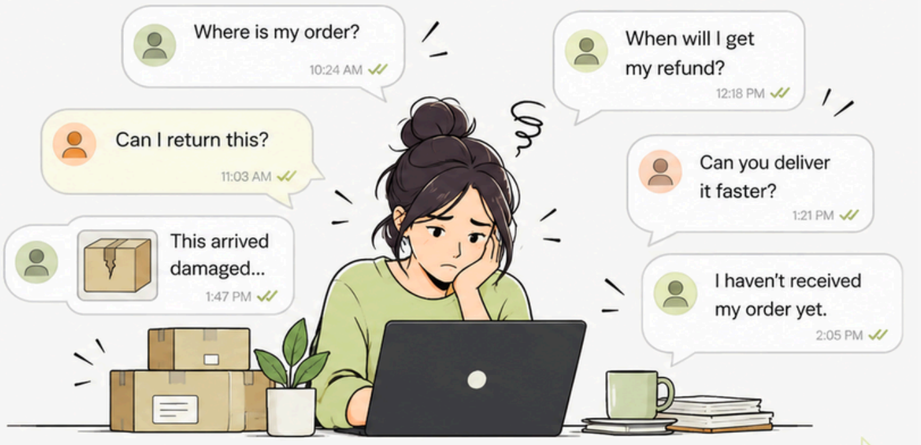
INVESTIGATE. DECIDE. ACT. VERIFY.



Team - Break The Bug

Member 1 - Avinash Singh Pal
Member 2- Saksham Upadhyaya
Member 3 - Riya Mehta

Problem Overview



Repetitive support requests leave small sellers with less time to run their business.

What makes support difficult?

- **Fragmented information :** Orders, payments, courier updates and customer conversations sit across different systems.
- **Complex fulfilment :** Delivery promises vary by PIN code, locality, warehouse and courier.
- **Inconsistent decisions :** Similar complaints may receive different outcomes.
- **Refund and return risks :** Genuine claims must be resolved without duplicate or unsupported refunds.
- **Poor traceability :** Checks, decisions and outcomes are difficult to track.

Why it matters?

Time : Repetitive queries take time away from core business operations.

Customer Trust : Delayed or inconsistent responses lead to customer frustration.

Financial exposure : Dedicated support staff may be unaffordable for small sellers.

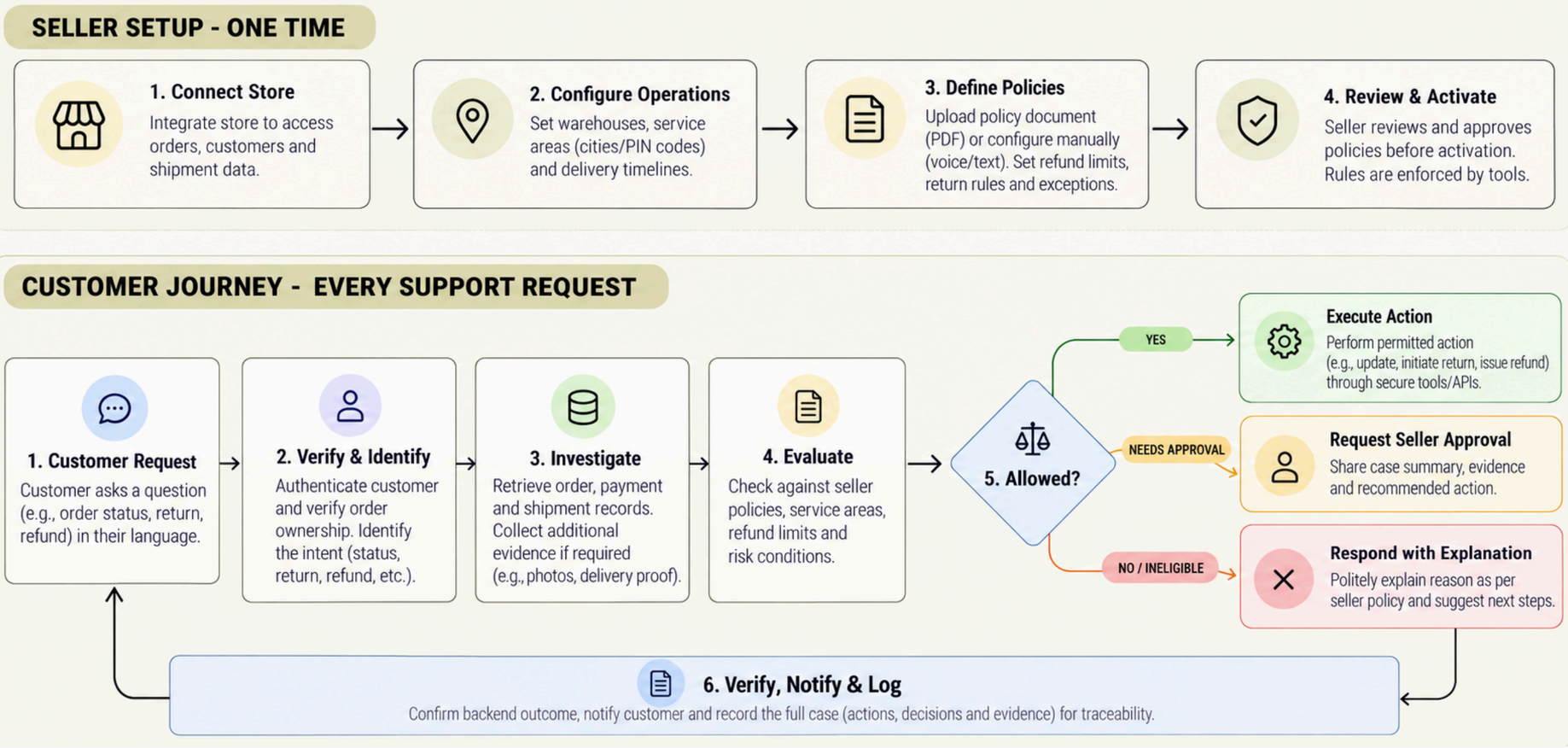
The Underlying Gap

Answering a query is not the same as resolving it.

Sellers need support that can investigate trusted records, apply their policies and safely complete permitted actions.

Proposed Solution

An autonomous, policy-aware support agent that investigates customer issues, executes authorized resolutions and verifies outcomes within seller-defined boundaries.



Core Capabilities

- **Context-aware investigation :** Tool-based order, payment and shipment retrieval.
- **Policy-bounded autonomy :** Seller-approved rules, refund limits and controlled execution.
- **Evidence-backed resolution :** Evidence collection, verified outcomes and human escalation.
- **Proactive exception monitoring :** Detect shipment delays and trigger customer updates.

System Architecture & Technology Stack



FRONTEND

- Tailwind CSS
- React 18
- TypeScript
- Server-Sent Events(real-time updates)

BACKEND

- FastAPI
- SQLite
- PyYAML (policy rules file) - httpx (tool and API calls) -
- APScheduler (proactive delay checks, follow-ups)
- LLM SDK with tool calling

LIBRARIES / TOOLS

- scikit-learn, Hugging Face
- indic-transliteration, sentiment analysis
- YAML policy configuration
- sentence-transformers

Innovation & Competitive Advantage

What makes the approach valuable?

More than a chatbot. It takes real actions.

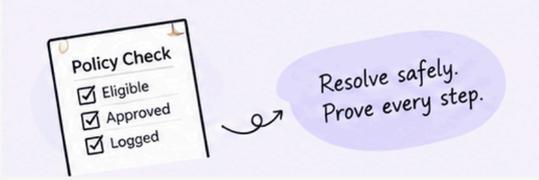
- **Acts, not just answers:** Checks the order, starts the return, issues the refund.
- **Follows the seller's rules**: Limits are written in plain language, like "refund up to ₹500".
- **Operational awareness:** Accounts for multiple warehouses, serviceable PIN codes and delivery zones.



Trust & Safeguards:

Every action is checked and recorded.

- **Policy-first execution:** Deterministic checks enforce eligibility, refund limits and approval requirements.
- **Checks before it acts:** Verifies the customer and blocks duplicate refunds or returns.
- **Seller stays in control:** Big refunds go to the seller with a short summary, and every action is logged.



How is it relevant?

Made for small sellers who answer the same questions all day.

- **Speaks your customer's language:** Detects the language and replies in the same one, including Hindi, Hinglish, English, and regional languages.
- **Handles everyday cases**: COD orders, return windows, and late deliveries.
- **Risk-proportional assessment:** Applies additional checks to incomplete, suspicious or conflicting claims.



Prototype to Real-World Deployment

FEASIBILITY

- **Core flow works end to end:** Chat, order lookup, policy check, then a return, refund, or escalation.
- **Simple, proven stack:** React, FastAPI, SQLite, an LLM with tool calling, and a YAML rules file.
- **Full demo workflow:** Customer chat, tool calls, seller dashboard, and action log.
- **Tested before the demo:** [X] of [Y] test messages pass, covering valid, denied, duplicate, and failed actions.

REAL-WORLD IMPACT

- **Faster replies:** Routine questions are answered in seconds, at any hour.
- **Less work for the seller**: Seller only sees the cases that need a decision.
- **Consistent decisions:** Every refund and return follows the seller's own rules.
- **Better for customers**: Clear updates and early warnings in their own language.
- **Demo result**: [X]% of test chats resolved without the seller.

SCALABILITY

Prototype | Hackathon

- Mock store and shipment data.
- Core workflow, policy checks and audit dashboard.

Operational Expansion | Next

- Multi-warehouse and PIN-code mapping.
- Evidence, risk checks and integrations.

Intelligent Scale | Future

- Proactive shipment alerts.
- Multi-channel and multi-seller support.
- Analytics and scalable infrastructure.

Constraint	Mitigation
Incorrect model decisions	Deterministic policy checks and validated outputs
Missing or conflicting data	Clear fallback and seller escalation
External API failures	Safe retries, idempotency and transaction-status reconciliation